No of additional sheets

LOURDES MATHA COLLEGE OF SCIENCE & TECHNOLOGY

LOURDES HILLS, KUTTICHAL P. O., THIRUVANANTHAPURAMM - 695 574

ANSWER BOOK

Name: Athira.c Class No: 13 Branch: MCA

Subject: Deep Learning Semester: Third

Mark Tabulation

Mark Divisions	Q.1	Q.2	Q.3	Q.4	Q.5	Q.6	Q.7	Q.8	Q.9	Q.10	Q.11	Q.12	Q.13	Q.14	Q.15
	3	2		2½	1	3	4	4							
Total															
Grand Total												20			

1) Sigmoid activation functions:

This is widely used on back propagation network because of the relationship between the value of the function at a part and the value of derivative at that part which reduces computational burden during training. sigmoid activation fuction refers that it is of two types as given below.

1) Binary sigmoidal function.

It perform input editing between 0 and 1 It is positive in nature. It is assigned bounded which mean the o/p cannot be less than 0 & more than that o/p which cannot

2) Bipolar sigmoid activation

$f(x) = \text{sigma}(x) = \frac{1}{1+\exp(-x)}$

2) Bipolar sigmoid function

It perform i/p editing b/w 1 & -1 - It can be +ve or -ve in nature. It is always bounded which means that its o/p cannot be less than -1 & more than 1

$f(x) = \text{sigma}(x) = \frac{2}{1+\exp(2x)} - 1 = \frac{1-\exp(x)}{1+\exp(x)}$

7) Back propagation ~~works~~

Back propagation is an algorithm for supervised learning of artificial neural networks using gradient descent. Given an artificial neural network and an error function the method calculates the gradient of the error function with respect to the neural network weights.

a) Initially the weights are assigned at random

b) The algorithm is iterates through main cycles of to process called forward phase and Backward phase.

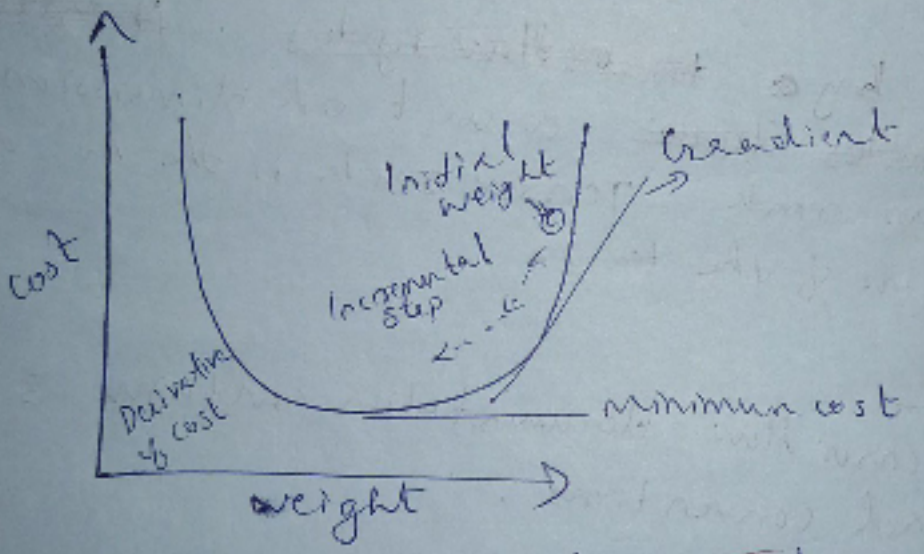
Forward phase

Neurons are activated in sequence from the input layer to the output layer. Applying neuron's weights and activation function along the way. an output signal is produced from the final layer.

Backward phase:

Output signal combined with the true target value. Difference in results ie; error is propagated backward in the network to modify the connection weights.

c) The technique used to determine how much a weight can be changed is known as Gradient Descent method.



At any point on the surface of the curve the gradient suggests how steeply the error will be reduced or increased for a change in the weight

Algorithm helps to change the weights that result in the greatest reduction in error.

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5) Tensor Ranks, shapes & Types

Tensor flow programs use a tensor data structure to represent all data

A tensor has a static type and dynamic dimensions

only tensors may be passed b/w nodes in the computation graphs

Ranks

In the tensor flow system, tensors are described by a ~~Tensor~~ ~~with Ranks~~. a unit of dimension by known as rank. Tensor rank is the no. of dimensions of the tensor.

Shape

Tensor flow documentation contains 3 notational conventions.

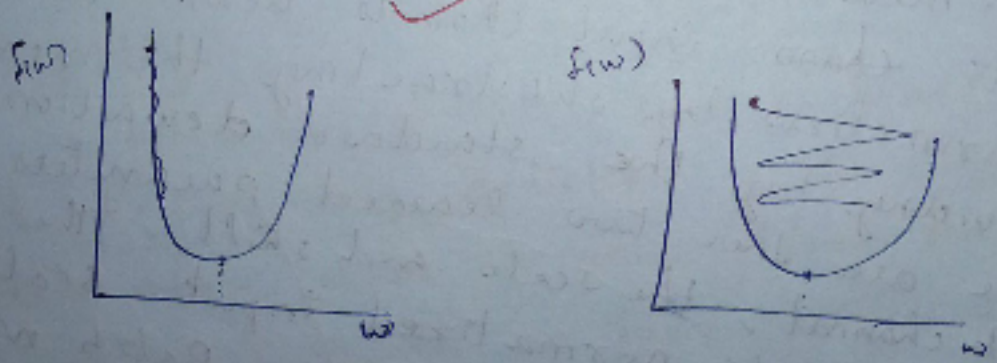
ADDITIONAL SHEET

2) Gradient descent algorithm

Gradient descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms it is basically used for updating the parameters of the learning model.

The goal of the gradient Descent algorithm is to minimize the given function.

1. Compute the gradient, the first order derivation of the function at that point.
2. Make a step in the direction opposite to the direction of the gradient



Types of gradient Descent

1. Batch Gradient Descent
2. Stochastic Gradient Descent
3. Mini batch gradient descent

21 Bipolar

1) Batch Gradient Descent

Use all training samples for one forward pass and then adjust weights

2) Stochastic Gradient Descent

Use one sample for a forward pass and then adjust weights

3) Mini batch gradient Descent

Use a batch of samples for a forward pass and then adjust weights

8) Batch Normalization

A batch normalization layer calculates the mean and standard deviation of each of its ~~them~~ input channels across the batch and normalizes by substracting the mean and dividing by the standard deviation. There are then two learned parameters for each channel, the scale and shift. The output is simply the normalized input, scaled by gamma and shifted by beta. Batch Normalizing is used to address the vanishing/ exploding gradients problems, and more generally the problem that the distribution of each layers inputs change during training as the

Parameters of the previous layers change. This technique consist of adding an operation in each layer simply zero centering and normalising the inputs, then scaling and shifting the result using two new parameters per layer.

Equations for batch Normalization

1. $\mu_B = \frac{1}{m_B} \sum_{i=}^{m_B} x^{(i)}$

2. $\sigma_B^2 = \frac{1}{m_B} \sum_{i=1}^{m_B} (x^{(i)} - \mu_B)^2$

3. $x^{(i)} = \frac{x(i) - \mu_B}{\sqrt{\sigma_B^2 + \varepsilon}}$

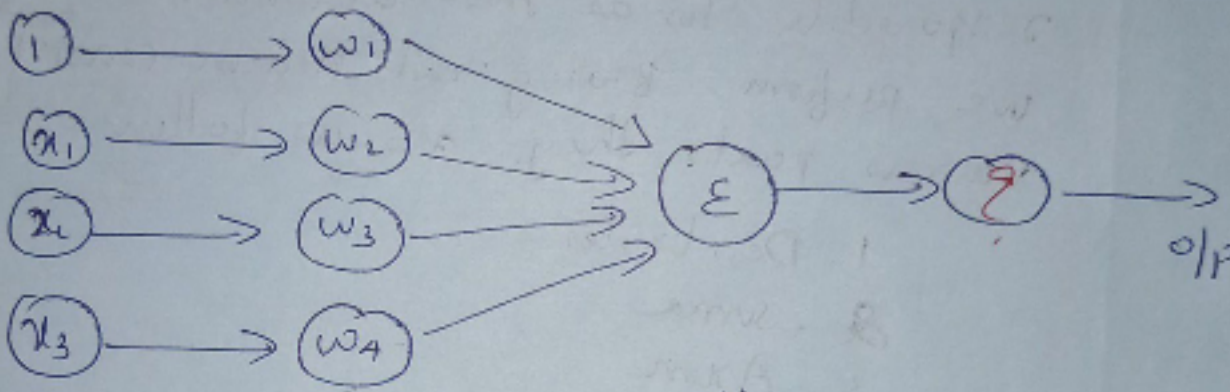
4. $z^{(i)} = \gamma x^{(i)} + \beta$

Overfitting

When the network tries to learn too much or too many details in the training data along with the noise from the training data which results in poor performance on unseen or test data set the when this happens the network fails to generalize

4) Perceptron Learning Algorithm

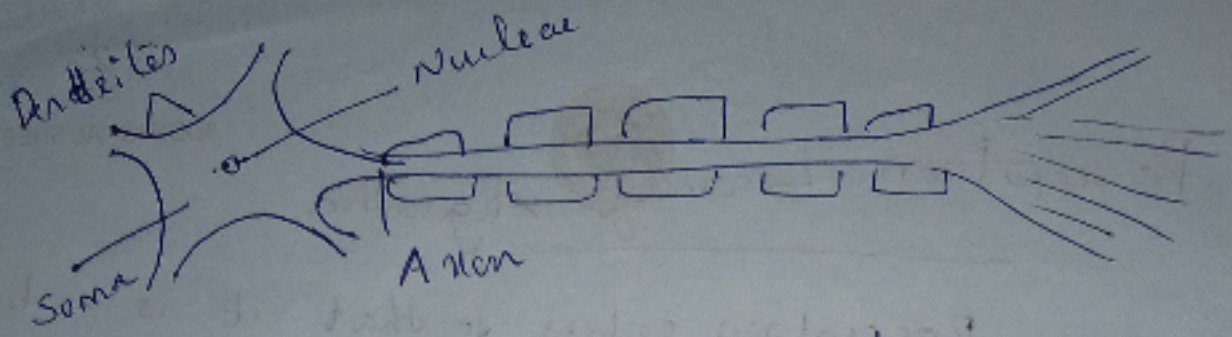
Perceptron refers to that it is a building block of an artificial neural network. Its basic component of perceptrons are as follows in the below input



Perceptron learning algorithm

P ← i/p with label 1;
N ← i/p with label 0;
initialize w randomly;
while 1 degree do
Pick random $x \in P \cup N$;
if $x \in P$ and $w_i x < 0$ then
$w = w + x$;
end
if $x \in N$ ~~and~~ $w_i x \geq 0$ then
$w = w - x$;
end
end

6)



Biological Neuron in which we can able to perform tasks. It is the process is actually present the human body which is responsible for as the actions and task that we perform. Biological neuron consist of various parts they are as follows

1. Dendrites
2. soma
3. Axon
4. Nucleus.

1) Dendrites: It are fibers which emanate from the cell body and provide the represented zones that ~~back~~ receive activation from other neuron.

2) Soma: the neuron is also known as cell body processes the incoming activities cell then into olp activation.

3) Axon: they are fibers acting as ~~transmission~~ transmission lines that send action to the other neuron.

The biological neuron where is used build the of the protein in the proven of keep the information in different form is the neuron it can be able to acce ss easy in the system.

4) Nucleus : It is contains the genetic material in the form of DNA exists is most type of cell.